

AKSHAJ MURHEKAR

 akshajmurhekar.github.io  akshajmurhekar  akshaj-murhekar  Google Scholar
akshaj.murhekar@utexas.edu

EDUCATION

The University of Texas at Austin

Aug. 2024 – May 2026

Master of Science in Information Science/Studies

GPA: 3.7/4.0

Specialization: Applied Machine Learning and Deep Learning

Selected Graduate Coursework: Deep Learning, Predictive Machine Learning, Accelerating Science with AI, Natural Language Processing, Statistical ML.

The University of Texas at Arlington

Aug. 2020 – May 2024

Bachelor of Science in Computer Science

GPA: 3.55/4.0

Selected Coursework: Artificial Intelligence, Operating Systems, Algorithms & Data Structures, Linear Algebra, Secure Programming.

MANUSCRIPTS & PRESENTATIONS

A. Murhekar, A. Mishra. *SYNAPSE: Neuro-Symbolic Visual Thought-to-Text Decoding via Topological Semantic Denoising*. Arxiv 2026. [PDF](#)

[Under review for EMNLP 2026]

- Introduced a lightweight neuro-symbolic framework that stabilizes neural text generation via inference-time topological graph purification and cross-modal exemplar retrieval.
- Eliminates end-to-end LLM fine-tuning while preserving biometric privacy by localizing raw EEG processing entirely within the encoder stack.

A. Murhekar, C. Liu, A. Mishra, S. Roychowdhury, J. Gwizdka. *SENSE: Efficient EEG-to-Text via Privacy-Preserving Semantic Retrieval*. ICMI 2026. [PDF](#)

Recipient of ACM SIGCHI Gary Marsden Travel Award. · [Master's Thesis Research]

- Architected a lightweight, ~6M parameter EEG-to-text framework that eliminates the need for expensive LLM fine-tuning by decoupling neural decoding into local semantic retrieval and zero-shot prompting.
- Engineered a privacy-preserving pipeline that performs semantic keyword extraction fully on-device, achieving generative quality that matches or surpasses state-of-the-art fine-tuned baselines while keeping raw neural data local.

S. Shukla, J. Torres, **A. Murhekar**, C. Liu, A. Mishra, J. Gwizdka, S. Roychowdhury. *A Survey on Bridging EEG Signals and Generative AI*. Arxiv 2025. [PDF](#)

[Under review for Neural Networks — Master's Thesis Research]

- Consolidated a structured review of literature across EEG-to-image, text, and audio synthesis, providing a foundational analysis of generative architectures, neural encoding strategies, and standardized benchmarks to support reproducible BCI research.

A. Murhekar, R. Tiwari. *Development of an Autonomous Medical Supply Cart to Streamline Nurse Workflow*. 2023 Biomedical Engineering Society (BMES) Annual Meeting, Seattle, WA. [Link](#)

[Poster Presentation]

RESEARCH EXPERIENCE

- UT Austin**, *Graduate Research Assistant* Aug. 2025 – Present
- **Thesis:** Developing a high-fidelity EEG-to-Text decoding framework leveraging LLMs for non-invasive neural signal translation.
 - Advisors: Abhijit Mishra, Shounak Roychowdhury, Jacek Gwizdka.
- UTA Research Institute (UTARI)**, *Undergraduate Research Assistant* 2023 – 2024
- Engineered data post-processing pipelines for autonomous drone systems utilized in large-scale 3D terrain mapping and aerial reconnaissance.
- UT Arlington, NSF REU**, *Research Fellow* 2023 – 2024
- Architected an automated text-to-video generative pipeline for synthetic educational content featuring synchronized voiceover and visual assets.
- UT Arlington, NSF REU**, *Research Fellow* 2022 – 2023
- Developed an autonomous person-following drone for the visually impaired; implemented on-device GANs to generate real-time musical alerts for obstacle avoidance.

TEACHING & MENTORSHIP

- UT Austin**, *Graduate Teaching Assistant* Spring 2025, Fall 2025
- **MKT 360: Information and Analysis**
 - **UGS 303: The Savvy Healthcare Consumer**
- UT Arlington**, *Undergraduate Academic Tutor* 2022 – 2024
- **Courses Tutored:** Data Structures & Algorithms, Artificial Intelligence, Operating Systems, Discrete Mathematics, Linear Algebra, and Calculus I–III.

SELECTED TECHNICAL PROJECTS

- Scientific Sycophancy Benchmark:** Engineered an evaluation pipeline uncovering an 85% failure rate where frontier LLMs prioritize helpfulness over physical laws, motivating a verification-aware RAG framework.
- KV Cache Optimization:** Engineered a PyTorch-based algorithm to reduce Llama-3 system memory usage by 50% during inference.
- Audio-Visual Source Separation:** Architected a U-Net with visual conditioning to separate target audio sources from mixtures via latent space modulation.
- LLM Fine-tuning (Review Feedback):** Fine-tuned Qwen2.5-1.5B using PEFT and Unsloth on a 46M-token synthetic dataset for high-throughput inference.

AWARDS & HONORS

- Presidential Scholarship** — Full-Tuition Merit Award (\$108,000) 2020 – 2024
A premier university-wide merit scholarship awarded to the top percentile of incoming students; provides comprehensive coverage of tuition and expenses for four years of undergraduate study.
- NIH Research Grant** — National Institutes of Health 2023 – 2024
Selected for NIH-funded development of autonomous healthcare robotics to optimize clinical nursing workflows.
- NSF REU Award (2x Recipient)** — National Science Foundation 2022 – 2024
Two-time recipient of the nationally competitive Research Experience for Undergraduates award.
2023: Funded for the design and implementation of an AI-based educational platform.
2022: Funded for the development and field-testing of an autonomous research-grade drone.

Undergraduate Research Appreciation Award — UTARI

2024

Recognized by the UTA Research Institute for technical excellence during a multi-semester research appointment.

Undergraduate Research Rank 3 — College of Engineering

2023

Ranked 3rd out of 33 competing REU teams for technical innovation and presentation of an autonomous drone system.

TECHNICAL SKILLS

Languages & ML: Python, PyTorch, Transformers, PEFT/LoRA, vLLM, Unsloth

Tools & Systems: Linux/Bash, Docker, Git, $\text{\LaTeX}$, Kubernetes, AWS, Nsight Compute